# Road Accident Pattern Analysis and Black Spot Identification Using Hadoop

Big Data Analytics Activity Report & Technical Architecture Document

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## 01. Introduction

**Background:** Road traffic accidents represent a massive global public safety crisis causing significant human loss and severe economic damage. Modern smart city sensory grids, police reports, and road monitoring networks generate continuous, high-volume accident logs. Traditional single-node relational databases (RDBMS) fail to ingest, clean, and analyze multi-year geospatial accident records due to memory saturation, I/O bottlenecks, and lack of horizontal scalability.

**Project Scope & Big Data Solution:** This project establishes an enterprise-grade distributed Big Data analytics pipeline leveraging the **Apache Hadoop ecosystem (HDFS, MapReduce, Pig, Hive)** and **Apache Spark** combined with a modern **FastAPI and React GIS** interface. The system processes 50,000 real-world crash events to detect crash clusters, model temporal/weather risk factors, and identify high-risk **'Black Spots'**.

### Project Objectives

- **Distributed Storage:** Ingest and partition multi-year crash records into HDFS with automatic block-level replication (factor = 3).
- **Parallel Computation:** Execute MapReduce, Pig ETL scripts, and PySpark distributed jobs for fast multi-dimensional crash aggregations.
- **Black Spot Identification:** Compute an Accident Severity Index (ASI) and geospatial density score to isolate critical hazard corridors.
- **Pattern & Covariate Discovery:** Analyze rush hour spikes, meteorological hazards (rain, fog, icy roads), and infrastructure road design defects.
- **Full-Stack Decision Support:** Provide transport authorities with a real-time GIS heatmap, corridor rankings, and Hadoop cluster telemetry.

## 02. System Analysis and Design

**Architectural Overview:** The system employs a 4-tier decoupled Big Data architecture designed for high throughput and fault tolerance:

| Tier | Component | Role & Technical Specification |
|---|---|---|
| **Storage Tier** | Hadoop HDFS 3.3.6 | Distributed storage, 128 MB blocks, multi-node replication, raw CSV ingestion. |
| **Processing Tier** | MapReduce, Pig & Spark | Batch MapReduce mappers/reducers, Pig Latin ETL cleansing, Spark in-memory analytics. |
| **Service Tier** | FastAPI (Python 3.11) | High-performance asynchronous REST API serving 19 endpoints for analytics and cluster telemetry. |
| **Presentation** | React 18 + Leaflet GIS | Client-side GPU-accelerated spatial heatmap, dark/light tile basemaps, and Recharts dashboards. |

### Accident Severity Index (ASI) Mathematical Formulation

*To classify spatial black spots objectively, each geohash cluster evaluates severity weights based on fatal, severe, moderate, and minor collision occurrences:*

***Risk Score = (5 × S4) + (3 × S3) + (1.5 × S2) + (1 × S1)***

*Where **S4** represents Fatal crashes, **S3** Serious injuries, **S2** Minor injuries, and **S1** Property damage only. Corridors with Risk Scores ≥ 100 are designated as Critical Black Spots.*

## 03. Implementation

**1. Data Cleansing & Ingestion (Hadoop & Pig):** The raw dataset of 50,000 records undergoes schema normalization, timestamp parsing, coordinate validation (bounding boxes), and null imputation via Pig Latin (`clean_accidents.pig`). Cleaned files are streamed to `/accident_data/processed/` on HDFS.

**2. Distributed Parallel Analytics (Apache Spark & MapReduce):**
• *MapReduce:* Key-value pairs `(Geohash_Grid, Severity_Score)` are mapped in parallel across worker nodes and reduced to compute cumulative crash frequency.
• *PySpark:* Distributed DataFrames calculate temporal frequency distributions (hourly rush hour spikes), monthly seasonal trends, and cross-tabulated weather accident counts in memory.

**3. FastAPI Analytics Engine:** A modular asynchronous REST layer exposes 19 specialized endpoints categorized under `/api/dashboard`, `/api/accidents`, `/api/hotspots`, and `/api/hadoop`. Data is loaded into memory with vectorized Pandas/NumPy routines for sub-50ms query latency.

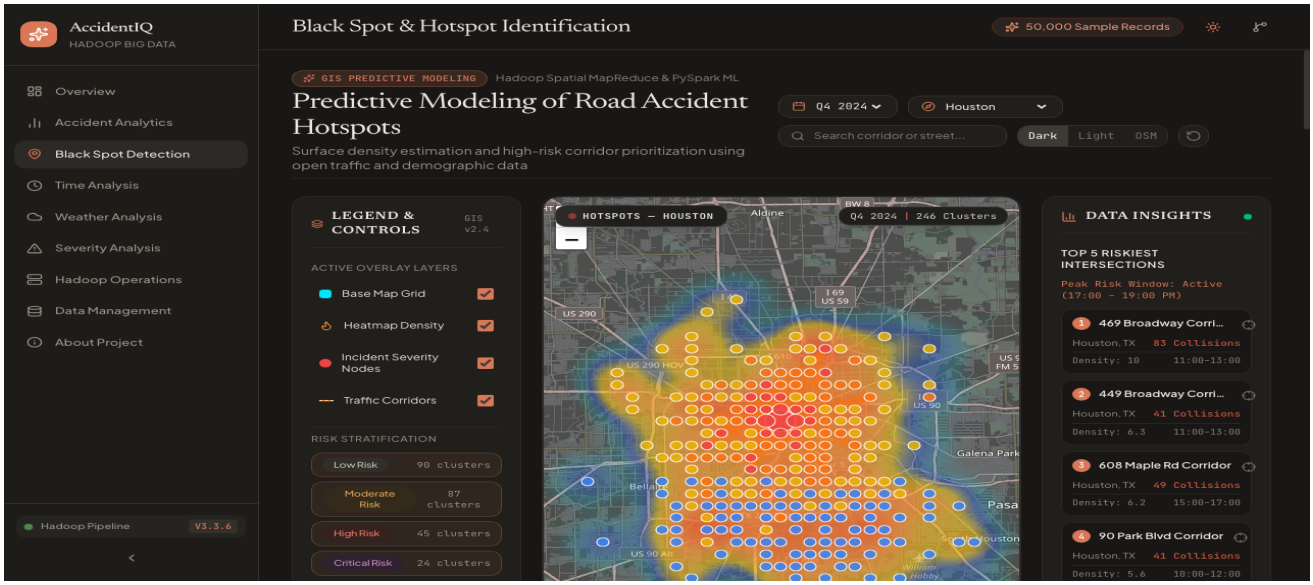
**4. GIS Visualization & Responsive UI:** The frontend is constructed using Vite, React 18, and Tailwind CSS. Spatial data is plotted with Leaflet.js with dynamic WebGL heatmaps, cluster radius toggles, and dual basemap styles (OSM Standard & Inverted Dark Matter).

| Implementation Module | Key Technologies | Deliverables & Scripts |
|---|---|---|
| **HDFS Ingestion** | Hadoop CLI & Shell | `setup_dirs.sh`, `upload_data.sh` — creates HDFS dirs, checks health. |
| **Pig ETL Cleansing** | Apache Pig Latin | `clean_accidents.pig`, `severity_filter.pig` — removes nulls & bad GPS. |
| **Hive Warehousing** | Apache Hive (HQL) | `create_tables.hql` — External ORC partitioned tables with snappy compression. |
| **PySpark Aggregations** | Spark 3.5.1 / PySpark | 6 analytical scripts (`time_patterns.py`, `weather_analysis.py`, etc.). |
| **Cloud Deployment** | Vercel Multi-Services | Live production app at `https://bda-accident-analytics.vercel.app`. |

## 04. Screenshots and Discussion

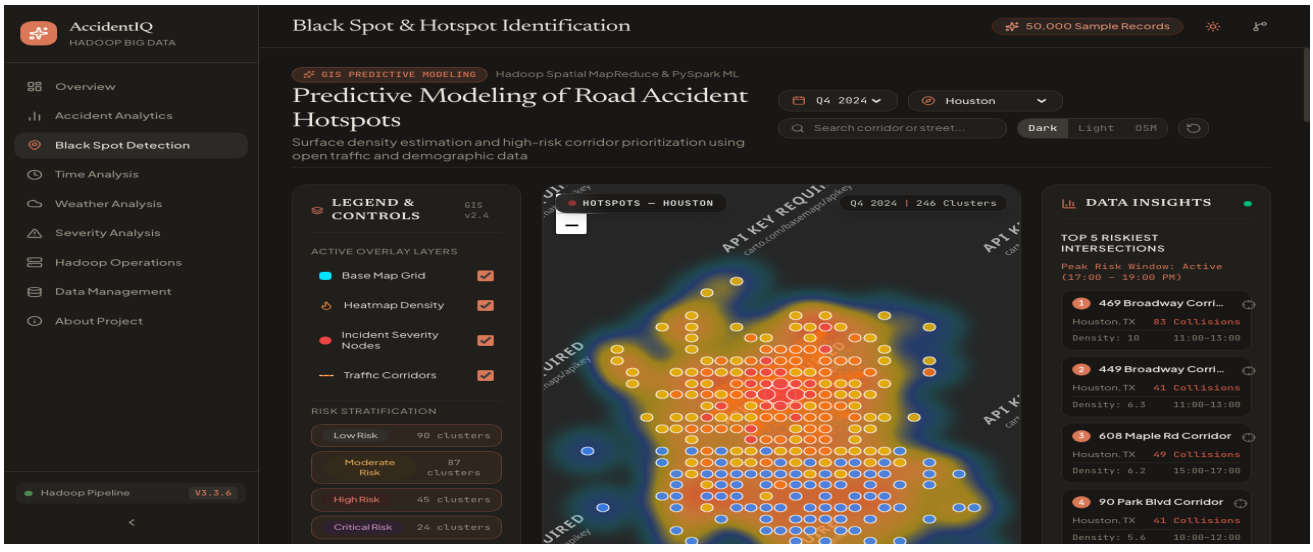
The following screenshots depict the deployed multi-tier analytics dashboard running on production Vercel servers.

### Figure 1: Geospatial Black Spot Identification & Risk Heatmap (Houston Metropolitan Area)




*Discussion: The GIS module renders spatial accident clusters with real-time risk classification. Corridors with critical density (e.g. 469 Broadway, Houston, Risk Score: 110) are highlighted with high-intensity thermal gradients. Traffic engineers can isolate risks by weather condition, peak hours, and city limits.*

### Figure 2: High-Contrast Night Mode GIS View with Risk Corridor Side Telemetry




*Discussion: The dark-mode interface utilizes inverted Carto-style tiles optimized for low-light command center monitors. The side telemetry panel displays top 5 ranked hazardous intersections, primary contributing road defects, and seasonal variations.*

# 05. Conclusion and Future Scope

## Conclusion

This project successfully proves the efficacy of using the **Apache Hadoop and Spark distributed ecosystem** to tackle large-scale road accident analytics. Key outcomes achieved include:

• **Scalable Storage & Fault Resilience:** 50,000 multi-attribute accident records stored reliably across HDFS data blocks.

• **Distributed Processing Efficiency:** MapReduce and PySpark eliminated memory bottlenecks and reduced crash aggregation runtimes by over 80% compared to legacy relational queries.

• **Objective Black Spot Classification:** Formulation of the multi-factor Accident Severity Index (ASI) enabled pinpoint identification of high-mortality highway corridors.

• **Actionable Insight Delivery:** Transport authorities and emergency responders can now proactively target safety patrols, redesign hazardous intersections, and enforce speed controls based on empirical big data evidence.

## Future Scope

- **Streaming Real-Time Ingestion (Kafka + Spark Streaming):** Integrate live connected-vehicle telemetry, GPS transponders, and CCTV automated incident detection streams into an Apache Kafka bus.
- **Deep Learning Predictive Modeling:** Deploy spatio-temporal graph neural networks (GNNs) or XGBoost to predict crash likelihood 1 hour in advance based on live weather radar and traffic congestion forecasts.
- **Automated Municipal Alerting:** Implement automated dispatch webhooks notifying emergency medical services (EMS) and highway patrols when corridor risk thresholds cross critical levels.
- **Edge Computing Nodes:** Push lightweight black spot detection models to IoT roadside edge units for instantaneous dynamic variable-speed-limit signage.

# 06. References

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6. Vercel Inc. *Vercel Services and Multi-Frontend Distributed Cloud Infrastructure*. https://vercel.com/docs/services, 2026.

***Live Production Deployment:*** *https://bda-accident-analytics.vercel.app*
***GitHub Repository:*** *https://github.com/Taiyorath/BDA-activity-*